

7 **(short)** A logic circuit has four inputs and five outputs. The four inputs A_1 , A_0 and B_1 , B_0 represent two 2-bit unsigned binary numbers. The five bits of the output C , i.e., C_4 , C_3 , C_2 , C_1 , and C_0 , are the result of the number A being raised to the power B . Derive the expressions for the output C of a calculator circuit to be implemented using NAND and inverter gates only. Note: $A^0 = 1$ for all values of A . [10]